II श्री गणेश प्रसन्न II

CAMPBELL-WALSH-WEIN · BOOK 10

Testis, Penis & Urethra

Concise rapid revision · facts answered in Papers 1–4 point to that answer

highlighted number to memorise · KEY POINT and ALGORITHM boxes carry the reasoning · P2 = covered in full in that Paper's answer book; the list under it gives what that answer misses

Chapter	Topic
Ch 78	Surgical, Radiographic and Endoscopic Anatomy of the Retroperitoneum
Ch 79	Neoplasms of the Testis
Ch 80	Surgery of Testicular Tumours
Ch 81	Minimally Invasive Retroperitoneal Lymph Node Dissection
Ch 82	Tumours of the Penis
Ch 84	Inguinal Node Dissection
Ch 83	Tumours of the Urethra
Ch 85	Benign Disorders of the Penis and Urethra
Ch 86	Urethral Stricture Disease
Ch 87	Surgery of the Scrotum and Seminal Vesicles
★	Putting it together

78 Surgical, Radiographic and Endoscopic Anatomy of the Retroperitoneum

1 · Definition, boundaries and organs

P1 Describe in brief the anatomy of the retroperitoneum… retroperitoneal laparoscopic nephrectomy — boundaries, primary vs secondary organs, flank layers, lumbodorsal fascia at 12th rib tip, three spaces, lateroconal fascia, perirenal cone open below, fat colour

Not in the paper answer:

- Primary organs lie in the perirenal space; secondary organs lie in the anterior pararenal space.
- Secondary organs exclude the first part of the duodenum (**D1**) and the pancreatic tail; the **anal canal** is primarily retroperitoneal.

> **KEY POINT**
> Secondarily retroperitoneal organs lie **in front of** the kidney, so colon, duodenum and pancreas are always in the surgeon's path to the kidney.

2 · Posterior abdominal wall and surface landmarks

- The upper pole is at the 11th rib (left) and 12th (right, lower). The hilum is at **L1**; the lower pole at L3–L4.
- **Psoas major** (T12–L5 to lesser trochanter) flexes the hip with iliacus. **Quadratus lumborum** lies posterolateral to psoas and stabilises the 12th rib; psoas minor is inconstant.
- Dividing the **costovertebral (lumbocostal) ligament**, which tethers the 12th rib to L1–L2, frees the rib in posterior approaches.
- The **intercostal bundle** lies under the rib above (vein, artery, nerve from top down), so the nerve is least protected. Supracostal access may cross the pleura and has more complications than subcostal.
- **Dorsal lumbotomy** incises the lumbodorsal fascia vertically, lateral to erector spinae and quadratus, so **no muscle is cut**.

3 · Retroperitoneal fasciae and spaces

Collections spread along these spaces.

Three fetal strata form three fasciae and spaces: outer (transversalis; posterior pararenal), intermediate (Gerota; perirenal, genitourinary) and inner (fusion fascia; anterior pararenal, gastrointestinal).

- The posterior Gerota lamina fuses medially with transversalis; this **must be incised to reach the hilum**.
- The Toldt (anterior) lamina is thinner than the Zuckerkandl (posterior) lamina.
- The perirenal cone may cross the midline below the hilum. Urinoma, haematoma and abscess are contained laterally but **track caudally**.
- The **anterior pararenal** space holds colon, pancreas and D2–D3, and is entered from the **white line of Toldt**.

4 · Gastrointestinal relations (operative hazards)

- **Right**: the pancreatic head lies in front of the inferior vena cava (IVC), and D2 lies on the right hilum, hence the **Kocher manoeuvre**. D3 crosses between the superior mesenteric artery (SMA) and the aorta.
- **Left**: pancreatic tail, spleen and stomach lie near the upper pole, and splenic vessels are at risk when the pancreas is lifted off renal fascia.

5 · Great vessels

Aorta and branches

The aorta enters at **T12**, left of the IVC, and bifurcates at **L4**.

Branch	Type	Level	Supplies
Inferior phrenic	Paired lateral	First, below hiatus	Diaphragm; superior adrenal
Celiac	Anterior	T12	Foregut
Middle adrenal	Paired lateral	SMA level	Adrenal
SMA	Anterior	L1–L2	Midgut
Renal	Paired lateral	L1–L2, below SMA	Kidney; inferior adrenal
Gonadal	Paired **anterolateral**	Below renals	Testis/ovary
Lumbar	Paired posterior	Four pairs, L1–L4	Wall, spine, cord
Inferior mesenteric (IMA)	Anterior	L3–L4	Hindgut

- Supernumerary renal arteries occur in ≥ ¼ (see Ch 84).
- **Ureteric supply**: proximal from renal; middle from aorta, common iliac and gonadal; distal from internal iliac. The abdominal ureter is supplied **medially**, the pelvic ureter **laterally**.
- IMA ligation is safe because the marginal artery of **Drummond** feeds the colon. **SMA ligation is not safe**: it causes severe bowel ischaemia.

IVC and veins

- The IVC passes the diaphragm at **T8**.
- The **left renal vein** crosses in front of the aorta below the SMA. It receives gonadal (right-angle entry), adrenal and ≥ 1 **lumbar vein** (avulsion risk). The **right** is short, without tributaries, and duplicated in one-sixth. Both enter the IVC at L1.
- The **right gonadal vein** enters the anterolateral IVC; the short **right adrenal vein** enters the posterolateral IVC.
- Left varicocele predominates (longer vein, perpendicular entry). A **new right varicocele** needs imaging for a retroperitoneal mass, e.g. renal cell carcinoma with venous thrombus.
- Dissect the right ureter **lateral** to the gonadal vein; medial dissection risks IVC injury and gonadal avulsion.
- The inferior mesenteric vein (**IMV**) lies in the left mesocolon and at reoperation is mistaken for the gonadal vein.

Variants (essential)

P1 P3 IVC anomalous development and retrocaval ureter; Retrocaval ureter — renal collar embryology, circumaortic/retroaortic vein, retrocaval ureter types and repair

- † A retroaortic left renal vein (≈ 2–3%) is injured when lumbars are divided or the aorta rolled in retroperitoneal lymph node dissection (RPLND).
- † A circumaortic vein (≈ 5–7%) has a posterior limb that mimics a lumbar vein, so check the pre-operative CT.
- † Retrocaval ureter is a **venous** anomaly with a "reverse-J" on pyelogram, repaired by dismembered pyeloplasty in front of the IVC.

6 · Lymphatics

P1 Lymphatic drainage of testis and scrotum, RPLND templates and penile cancer implications — right- to-left cephalad flow, cisterna chyli L1–L2, three nodal regions, landing zones (right interaortocaval, left para-aortic), right-to-left crossover only

- Lumbar nodes are the primary drainage of kidney, adrenal, ureter and gonad.

> **KEY POINT**
> The testis is embryologically retroperitoneal, so it drains to where it began. Crossover is one-way (right to left), so right templates include interaortocaval and some para-aortic tissue. See Ch 80, 81.

7 · Nerves

Autonomic

- **Parasympathetic** outflow is craniosacral (III, VII, IX, X, S2–S4). **Sympathetic** outflow is thoracolumbar, **T1–L2**.
- **Sympathetic trunks** lie medial to psoas behind the great vessels, and are met when lumbar vessels are divided.
- **Ejaculation**: † postganglionic fibres from the **T12–L3** ganglia cross in front of the aorta and IVC and converge in the hypogastric plexus below the IMA. They drive **emission** and **bladder neck closure**, so extensive dissection causes anejaculation or retrograde ejaculation, while erection is spared. See Ch 81.

Lumbar (lumbosacral) plexus

Nerve	Roots	Course / motor	Sensory
Subcostal	T12	Below 12th rib	Skin over hip
Iliohypogastric	L1	Over quadratus; internal oblique, transversus	Gluteal, pubic
Ilioinguinal	L1	As above	Upper medial thigh, penile root, anterior scrotum
Genitofemoral	L1–L2	**On anterior psoas**; genital branch to cremaster	Anterior scrotum; upper anterior thigh
Lateral cutaneous thigh	L2–L3	Sensory only	Lateral thigh
Obturator	L2–L4	Medial to psoas, to obturator canal; adductors	Medial thigh
Femoral	L2–L4	Lateral to psoas; quadriceps	Anterior thigh, medial leg

Nerve	Setting	Result of injury
Genitofemoral	Psoas hitch, laparoscopic varicocelectomy, RPLND	Anterior scrotal/thigh numbness
Obturator	Lateral-wall TURBT (bladder tumour resection); pelvic node dissection	Adductor jerk causing bladder perforation
Femoral	Retractor blade on psoas; lithotomy	Loss of knee extension
Hypogastric plexus	Extensive retroperitoneal dissection	Anejaculation/retrograde ejaculation

8 · Imaging anatomy

- On axial CT the **duodenum lies against the right hilum** and the left renal vein in front of the aorta.
- † Before RPLND or nephrectomy, check accessory renal arteries, retro-/circumaortic left renal vein and nodal size by zone.

79 Neoplasms of the Testis

1 · Epidemiology and risk factors

- Over 95% are **germ cell tumours (GCTs)**: seminoma or non-seminomatous GCT (NSGCT); 5% are extragonadal. It is the commonest solid malignancy in men aged **20–40 y**.
- Cisplatin raised metastatic cure to **80–90%**, so the aim is **minimum toxicity without losing curability**.
- Bilateral in **2%**, mostly metachronous.
- **Risk factors**: White race, cryptorchidism, family history, personal history, germ cell neoplasia in situ (GCNIS) and infertility. Undescended testis (UDT) has a relative risk (RR) of **4–6**, falling to 2–3 with orchidopexy before puberty. A previous GCT gives a 12-fold contralateral risk, yet the 15-y cumulative incidence is only 2%.
- **GCNIS** is an arrested fetal gonocyte formed **before birth**. It lies beside invasive GCT in 80–90% and invades in **50% at 5 y** and 70% at 7 y. Contralateral GCNIS is found in 5–9%, and in ≈36% with atrophy, UDT or age <40.
- Microlithiasis alone needs no work-up; it matters only after a previous GCT.

> **KEY POINT**
> GCT is a disease of the arrested gonocyte: UDT, family history and atrophy mark an abnormal developmental field, so orchidopexy reduces but never abolishes risk.

2 · Pathology and classification

P3 Classify testicular tumors and describe management of seminoma — WHO 2016 groups, i(12p), any NSGCT element = NSGCT, seminoma IHC

Not in the paper answer:

- Isochromosome 12p, i(12p), proves GCT origin in a carcinoma of unknown primary and in somatic-type malignancy from teratoma.

Subtype	Markers	IHC / histology	Hallmark
Seminoma	Human chorionic gonadotrophin (hCG) ≈15%; **never alpha-fetoprotein (AFP)**	CD117+, CD30−, placental alkaline phosphatase (PLAP)+	4th–5th decade; can transform to NSGCT
Spermatocytic	None	OCT3/4−, PLAP−; three cell types	6th decade; never mixed, not with UDT; **benign** unless sarcomatous/anaplastic
Embryonal carcinoma (EC)	AFP/hCG or normal	AE1/AE3+, OCT3/4+, CD30+†, c-KIT−	Totipotent; spreads with normal markers; predicts occult spread
Yolk sac	AFP, not hCG	Schiller–Duval bodies (≈50%), hyaline globules	Pure form: children, mediastinum
Choriocarcinoma	Very high hCG	Syncytio- and cytotrophoblasts	Spreads by blood (lung, liver, brain); bleeds; poor risk
Teratoma	Normal/mild AFP	≥2 germ layers; maturity irrelevant after puberty	Benign histology, malignant genetics; **chemoresistant**

- Teratoma is present in ≈half of mixed GCTs. Unresected, it causes growing teratoma, somatic transformation or late relapse, so surgery is integral to cure.
- hCG shares its α-subunit with TSH and LH, so it can cause hyperthyroidism and gynaecomastia.

> **KEY POINT**
> Management follows the most chemoresistant element: EC drives occult metastasis, choriocarcinoma haemorrhage, teratoma surgery.

3 · Presentation

- A **painless mass** is commonest. Metastases are present at diagnosis in ≈2/3 of NSGCT and 15% of seminoma; 10–20% present with metastatic symptoms such as back pain or a left supraclavicular node.
- About 1/3 are first misdiagnosed, so re-examine epididymo-orchitis 2–4 weeks after antibiotics.

4 · Diagnosis and initial management

Scrotal ultrasound (US)

- GCT is **hypoechoic**: homogeneous suggests seminoma, heterogeneous suggests NSGCT.
- In metastatic GCT with normal palpation, US seeks a **burned-out primary**, which still needs orchidectomy; normal testes mean extragonadal GCT.
- An impalpable <10 mm lesion with normal markers is GCT in 20–50%: options are orchidectomy, inguinal exploration with frozen section, or serial US.

Serum tumour markers

P1 Tumour markers in urology — sources, never-raised-in, half-lives, S stage, raised AFP = NSGCT

Not in the paper answer:

- Half-lives: AFP **5–7 days**, hCG **24–36 h**, LDH 24 h.
- After orchidectomy, a slow decline with negative imaging is monitored, but **rising markers mean chemotherapy**.
- An unstable patient with very high AFP/hCG gets chemotherapy first and orchidectomy later.

False positives (hepatic disease, hypogonadal LH cross-reactivity, marijuana) occur, and miR-371a-3p is a GCT-specific marker not raised by teratoma.

Radical inguinal orchidectomy

- Remove testis and cord to the internal ring within 1–2 weeks. **Trans-scrotal approach or biopsy is contraindicated**. Orchidectomy alone cures 80–85% of clinical stage (CS) I seminoma and 70–80% of CS I NSGCT.
- **Testis-sparing surgery** suits synchronous bilateral or solitary testis (<2–3 cm, adequate androgen) or a suspected benign lesion, with frozen section; GCNIS then gets ≥20 Gy.
- Biopsy the contralateral testis if atrophy, UDT, age <40 or suspicious US.

5 · Staging

Spread and imaging

P1 Lymphatic drainage of testis and scrotum, RPLND templates and penile cancer implications — retroperitoneum first site 70–80%, landing zones, crossover

Not in the paper answer:

- CT abdomen–pelvis for all misses disease in **25–35% of CS I NSGCT**, so landing-zone nodes of 5–9 mm are suspicious. Equivocal nodes with normal markers are re-imaged in 6–8 weeks.
- Brain imaging if hCG >10,000 IU/L; no routine bone scan or FDG-PET at diagnosis.
- Chest CT if markers raised, abdominal disease, abnormal X-ray, or CS I NSGCT with LVI/EC.

Base decisions on CT and markers within 4 weeks of treatment, and offer **sperm cryopreservation** before RPLND, chemotherapy or radiotherapy.

TNM staging — the only TNM with a serum (S) category

Category	Definition
pTis	GCNIS
pT1	Testis/epididymis, **no LVI**; may invade tunica albuginea
pT2	LVI, or tunica vaginalis
pT3 / pT4	Spermatic cord / scrotum
N1 / N2 / N3	≤2 cm / >2–5 cm / >5 cm (pN1 ≤5 nodes, pN2 >5 or extranodal; pN invalid after chemotherapy)
M1a / M1b	Non-regional nodes or lung / other viscera
S1	LDH <1.5×N and hCG <5000 mIU/mL and AFP <1000 ng/mL
S2	LDH 1.5–10×N or hCG 5000–50,000 or AFP 1000–10,000
S3	LDH >10×N or hCG >50,000 or AFP >10,000

Stage	Criteria
IA / IB	pT1 / pT2–4, N0 M0 S0
IS	Any pT, N0 M0, **S1–3** (raised markers, no visible disease)
IIA / IIB / IIC	N1 / N2 / N3, M0, S0–1
IIIA	M1a, S0–1
IIIB	N1–3 or M1a with S2
IIIC	N1–3 or M1a with S3, or **any M1b**

IGCCCG risk — metastatic disease

Risk	NSGCT	Seminoma
Good	Testis/retroperitoneal primary, no NPVM, AFP <1000, hCG <5000, LDH <1.5×N (56%; 5-y survival 92%)	Any primary, no NPVM, any hCG/LDH, normal AFP (90%; 86%)
Intermediate	Any of AFP 1000–10,000, hCG 5000–50,000, LDH 1.5–10×N (28%; 80%)	**NPVM** (10%; 72%)
Poor	**Mediastinal primary**, NPVM, AFP >10,000, hCG >50,000 or LDH >10×N (16%; 48%, now ≈71%)	**None**

NPVM = non-pulmonary visceral metastases.

KEY POINT

IGCCCG uses markers at the start of chemotherapy, not pre-orchidectomy values, and is invalid at relapse. Seminoma risk depends on NPVM alone; mediastinal NSGCT is always poor risk.

6 · Treatment principles

- Centralise care, give full-dose cisplatin and resect all residual disease.

	Seminoma	NSGCT
CS I / II / III	85 / 10 / 5%	≈1/3 each
Occult metastasis CS I	10–15%	25–35%
Radiosensitive	Exquisitely	No (except brain)
Metastatic teratoma	Rare	Frequent, so surgery needed
Markers	Rarely useful	Central

- Seminoma relapses contain NSGCT in 10–15%, so biopsy non-responders.

GCNIS management

- Options are orchidectomy, radiotherapy **≥20 Gy** (lower doses allow recurrence) or observation. Up to 40% later need testosterone after radiotherapy.
- With a normal contralateral testis and a wish for paternity, orchidectomy is preferred because scatter harms the other testis. Cisplatin only partly clears GCNIS.

7 · NSGCT management

Every CS I option cures ≈100%; the choice is about toxicity.

CS I NSGCT

- Adjuvant therapy overtreats 70–80%.
- **Risk factors** are **LVI** (strongest) and **EC predominance**. High-risk tumours relapse in ≈50%, low-risk in <20%.
- **Surveillance**: relapse ≈28%, >90% within 2 y and mostly retroperitoneal; relapses are chemo-naive and curable.
- **Primary RPLND** removes teratoma and finds pathological stage II in 19–28%, mostly cured by surgery alone. pN1 is observed; pN2–3 (relapse >50%) gets **2 cycles adjuvant** (relapse ≤1%) or observation.
- **Bleomycin–etoposide–cisplatin (BEP) ×1** gives the lowest relapse (≈2–3%) but leaves teratoma, exposes all to late toxicity, and relapses are chemoresistant.
- Full bilateral nerve-sparing template (ejaculation >90%) is standard; a modified template is acceptable for clinically negative nodes.

ALGORITHM

1. Assess CS I NSGCT for LVI and EC predominance.
2. If low risk (neither), then surveillance.
3. If high risk, then surveillance, BEP×1 or expert nerve-sparing RPLND, chosen by preference and toxicity.

> **KEY POINT**
> Surveillance maximises quality-adjusted survival when relapse risk is <≈1/3. All options cure ≈100%; they differ in overtreatment versus difficulty of salvage.

CS IS, IIA–B

- **CS IS** is treated with induction chemotherapy.
- **CS IIA–B** with raised markers or nodes **>3 cm** gets chemotherapy, with or without post-chemotherapy RPLND (PC-RPLND). With normal markers and nodes <3 cm, **primary RPLND** is used: 13–35% are pN0, ≈30% have teratoma, and half avoid chemotherapy.

CS IIC–III: induction by IGCCCG

P2 CRE & BEP — BEP×3/EP×4 good, BEP×4 (VIP if lungs poor) intermediate/poor, toxicities

Not in the paper answer:

- BEP×3 equals BEP×4; etoposide–cisplatin (EP) ×3 and carboplatin are inferior.
- Second line is TIP×4 (paclitaxel, ifosfamide, cisplatin), VeIP×4 or VIP; TIP is best for late relapse.
- High-dose chemotherapy (HDCT) failed in three randomised trials, so it is for specialised centres.

Post-chemotherapy residual masses (NSGCT)

- With normal markers and a residual mass **>1 cm**, do **PC-RPLND** with a full bilateral template. Histology is necrosis in ≈40%, teratoma in ≈45% and viable tumour in ≈15%.
- No model, imaging or **FDG-PET** predicts necrosis (teratoma is not avid), so always resect.
- After a complete response (≤1 cm), observe, or resect if pre-chemotherapy nodes were >3 cm (controversial).
- With multiple sites, resect all, retroperitoneum first, because its necrosis predicts necrosis elsewhere. Somatic transformation (6–8%) needs complete resection.
- After viable tumour, adverse factors are intermediate/poor risk, incomplete resection and >10% viable cells.

Relapsed NSGCT

P1 Growing teratoma syndrome — enlarging mass, normal markers: stop chemotherapy, resect

- **Chemo-naive relapse** gets induction by IGCCCG, with cure >95%.
- **Early relapse (≤2 y) after chemotherapy** gets TIP/VeIP/VIP×4 or HDCT, then post-salvage surgery for residual masses.
- **Desperation surgery** is for rising markers with a single resectable site; survival is 33–57%.
- **Late relapse (>2 y)** occurs in ≈3%. Yolk sac tumour is commonest and chemoresistant, so **complete resection** is the treatment.

8 · Seminoma management

P3 P2 Classify testicular tumors and describe management of seminoma; Describe anatomy of lymphatic drainage of testis and scrotum. Describe management of stage one seminoma of testis. — CS I options table, risk factors, Calvert, stage II–III, residual mass, follow-up

Not in the paper answer:

- CS I risk factors are tumour **>4 cm** and rete testis invasion (relapse 12/16/32% with 0/1/2), but they are not validated, so surveillance is preferred for all.

- IIA–B nonbulky (≤3 cm): dog-leg radiotherapy 25–30 Gy with a 5–10 Gy boost, BEP×3/EP×4, or primary RPLND (SEMS, COTRIMS). Bulky or multiple nodes get chemotherapy.
- Carboplatin ×2 gives relapse ≈1.5%. Seminoma relapses late (median 14 months), and radiotherapy carries second malignancy risk.

Residual seminoma masses are necrosis in ≈90%, and many regress over 12–18 months.

ALGORITHM

1. If the residual seminoma mass is <3 cm, then observe.
2. If >3 cm, then FDG-PET ≥6 weeks after chemotherapy (earlier scans give false positives).
3. If PET-negative, then observe.
4. If PET-positive, then template PC-RPLND (or repeat PET/observe). Mass-only excision gives 33% local relapse.

- Relapse after radiotherapy gets first-line chemotherapy; after chemotherapy, VeIP×4 or HDCT.

9 · Brain metastases

- Present in ≈1% at diagnosis, with **choriocarcinoma or very high hCG**; they can bleed fatally. Give BEP×4, then resect residuals.

10 · Treatment sequelae and follow-up

- RPLND can cause ejaculatory loss (>80% after full template without nerve sparing). Orchidectomy alone causes hypogonadism in 10–20%.
- Chemotherapy toxic death is 0–2.4% (good risk) and 3–4.4% (intermediate/poor). Azoospermia is near universal; ≈half recover by 2 y, 80% by 5 y.
- Late effects: Raynaud 20–45%, neuropathy 14–43%, ototoxicity, hypogonadism 15–40%, **cardiovascular disease** and **second malignancy** (60–100% excess after cisplatin or radiotherapy, 200% after both).
- Follow-up is most intensive in years 1–2 (NSGCT) or 1–3 (seminoma).

KEY POINT

Because cure is expected, late cardiovascular disease and second cancers, which rise with each added modality, decide between equally curative options.

11 · Non-germ-cell tumours†

- †**Leydig cell**: the commonest sex cord–stromal tumour, causing precocious puberty or gynaecomastia. ≈10% are malignant; these resist chemo- and radiotherapy, so nodes need RPLND. A small marker-negative lesion suits testis-sparing excision.
- †**Sertoli**: mostly benign; the large-cell calcifying variant occurs with Carney complex and Peutz–Jeghers.
- †**Lymphoma**: the commonest testicular tumour in men **>50–60 y** and the commonest bilateral one. It is treated by orchidectomy, then R-CHOP with CNS prophylaxis and contralateral testis radiotherapy.

12 · Landmark trials

Trial	Design	Result	Take-home
IGCCCG 1997	Pooled 5202 NSGCT, 660 seminoma	Markers, NPVM, mediastinal primary predict outcome	Risk groups select regimen
PVB vs BEP	RCT, advanced	Equal survival, less neurotoxicity	BEP standard
BEP×3 vs ×4	RCT, good risk	Equal survival	BEP×3
VIP vs BEP×4	RCT, intermediate/poor	Equal, more toxicity	VIP only if lungs at risk
MRC TE19/EORTC 30982	Carboplatin ×1 vs radiotherapy, CS I seminoma	5-y RFS 94.7% vs 96%	Equivalent
MRC 20 vs 30 Gy	RCT, CS I seminoma	Equal relapse, less toxicity	20 Gy PA
Albers (German)	BEP×1 vs unilateral RPLND, CS I NSGCT	2-y PFS 99% vs 92%	Favours BEP×1
Williams	Adjuvant ×2 vs observation, pathological stage II	Relapse 6% vs 49%; same survival	Both acceptable
MRC surveillance	2 vs 5 CTs, CS I NSGCT	No difference	Fewer scans safe

RCT = randomised controlled trial; RFS = relapse-free survival; PFS = progression-free survival; PA = para-aortic.

80 Surgery of Testicular Tumours

1 · Evaluation of the testis mass

It is urgent, because delay escalates curative therapy.

- Examine supraclavicular nodes, breasts, abdomen and scrotum. Scrotal ultrasound (US) mainly checks the **contralateral testis** (bilateral synchronous ≈ **1%**). Markers (AFP, β-hCG, LDH) give diagnosis, stage and post-orchidectomy baseline.
- Discuss a prosthesis and † offer sperm banking.

2 · Radical inguinal orchidectomy

Treatment of choice: **early cord control** before handling the testis, removing testis, epididymis and cord to the internal ring. † A scrotal approach breaches a second (inguinal) lymphatic field and leads to local recurrence.

ALGORITHM

1. Make a 3–5 cm incision over the canal and open the external oblique.
2. **Preserve the ilioinguinal nerve**.
3. Mobilise the cord at the pubic tubercle and apply a **Penrose early clamp**.
4. Deliver the testis and divide the gubernaculum.
5. Dissect the cord to the internal ring.

- **High ligation**: ligate vas and gonadal vessels **separately** with **non-absorbable** suture, leaving a **1–2 cm** tail to mark the stump for excision at retroperitoneal lymph node dissection (RPLND).

KEY POINT

† **Scrotal violation**: inguinal exploration, high cord ligation and excision of the scrotal scar. Routine hemiscrotectomy is unnecessary; chemotherapy-era outcomes are not compromised.

Partial orchidectomy (testis-sparing surgery)

P4 Testis sparing surgeries for CA testis — indications (solitary/bilateral, small, polar), intraoperative US, cold ischaemia <30 min, frozen section, GCNIS 18–20 Gy, recurrence: completion orchidectomy

Not in the paper answer:

- Needs **all three**: polar, **≤2 cm**, contralateral testis compromised/absent. **Never elective with a normal contralateral testis.**
- Masses **<5 mm** with negative markers have low malignancy, so selective surveillance is an option. Cold ischaemia can be omitted if resection takes **<30 min**.
- GCNIS options are completion orchidectomy, **18–20 Gy** or observation (local recurrence ≈ 7.5%). **Local recurrence always needs completion orchidectomy**.

Delayed orchidectomy

Life-threatening metastatic germ cell tumour (GCT) gets chemotherapy first, then **delayed orchidectomy in all**, because the testis is a sanctuary with **discordant response**. An apparent extragonadal retroperitoneal primary shows intratubular neoplasia in up to 42%, so lateralising nodes mean orchidectomy on that side.

3 · RPLND — terminology and indications

Term	Indication
Primary RPLND	High-risk clinical stage (CS) I or low-volume stage II non-seminomatous GCT (NSGCT), **normal markers**
Post-chemotherapy (PC-RPLND)	After induction; residual **>1 cm**, normal markers
Salvage PC-RPLND	After second-line chemotherapy
Desperation	Markers still elevated; resectable
Redo	Retroperitoneal recurrence after prior RPLND
Late relapse	**>24 months** after complete response

Each step down means more chemoresistant disease and harder surgery.

Preoperative planning

- CT within **6 weeks** and markers within **7–10 days**; note vascular variants; no bowel prep.
- After **bleomycin**, test lung function and ask anaesthesia to use **low FiO_2, restrictive fluids**.
- Total IVC occlusion by thrombus needs inferior vena cava (IVC) resection; partial needs reconstruction.

4 · Templates

P1 Lymphatic drainage of testis and scrotum, RPLND templates and penile cancer implications — landing zones, bilateral infrahilar and modified templates, omit suprahilar/interiliac, 3–23% out-of-template, PC-RPLND full bilateral; modified only if ≤5 cm in landing zone + normal markers + good/intermediate risk; L1–L4 fibres to hypogastric plexus

- The bilateral template is bounded by renal vessels, ureters and common iliac arteries. A modified post-chemotherapy template needs a lesion **≤5 cm**.

5 · Open RPLND technique

- The approach is **transperitoneal** or **extraperitoneal** (less ileus).
- Incise from the caecum to the inferior mesenteric vein (IMV) and along the line of Toldt to the foramen of Winslow, freeing duodenum and pancreas. Use **liberal clips** to prevent chyle leak.
- **Split and roll**: split at 12 o'clock on the aorta from the left renal vein down to the inferior mesenteric artery (IMA), roll packets off and ligate lumbars.
- Lumbar arteries are usually 3 pairs, with the **2nd pair at IMA level**. The **right sympathetic trunk**, posterolateral to the IVC, is mistaken for lymphatic tissue.

Nerve sparing

- Left fibres run along the lateral aortic border near the IMA. Right fibres emerge **beneath the IVC** and cross the aorta obliquely, so they are at risk when lumbar veins are ligated.

> **KEY POINT**
> Identify fibres **prospectively, before dividing anything**; loop them and pass the packet beneath. Avoid traction, which causes neuropraxia and transient anejaculation.

Postoperative care

- No routine drain; if one is used, remove it at **<100 mL/24 h** on diet. No nasogastric (NG) tube, low-fat diet early, and home on days 3–5.

6 · Auxiliary procedures (PC-RPLND 23–45%)

- **Nephrectomy** is commonest, usually **left**. The strongest predictor is a residual mass **>10 cm**.
- **Vascular** repair is planned in every PC-RPLND, using a patch or graft.
- **IVC**: narrowing **<25%** is repaired by venorrhaphy, >25% by patch or graft; ligation is acceptable. **The aorta is never ligated**.
- **Liver**: histology is discordant in ≈half; resect, or ablate if **<3 cm** and unresectable. **Thorax**: do RPLND first, then resect lesions >1 cm unless the retroperitoneum shows fibrosis only (thoracic necrosis then in ≈89%).

7 · Decision making

- **Complete remission (≈70%)**: generally observe, though ≤1 cm residuals hold teratoma in ≈ 1/5 and viable tumour in ≈ 1/20.
- **Necrosis is predicted by** no teratoma in the primary, normal pre-chemotherapy AFP/β-hCG, raised LDH, >50% shrinkage and residual <2 cm.
- **After primary RPLND**: pN0, teratoma only or pN1 are observed; pN2 gets 2 cycles or observation. Observation cures ≈70%.

8 · Histology of residual mass after chemotherapy

Finding	Frequency	Outcome/action
Fibrosis/necrosis	**≈40%**	Relapse-free (RFS) and cancer-specific survival (CSS) ≈95%; no adjuvant
Teratoma	**≈45%**	RFS 80–90%; must be resected
Viable GCT	**≈15%**	Survival 50–70%; consider 2 adjuvant cycles

> **KEY POINT**
> **Resect every NSGCT residual >1 cm**: no imaging (including PET †) or marker separates necrosis from teratoma, and teratoma is chemoresistant, grows, relapses late and transforms. Surgery is both staging and the only treatment.

- **Fizazi** model for viable GCT: incomplete resection, ≥10% viable cells and IGCCCG intermediate/poor risk. With 0, 1 or ≥2 factors, 5-y overall survival (OS) is 100/83/51%.
- Adjuvant chemotherapy: NCCN for incomplete resection or intermediate/poor risk; EAU for incomplete resection only. Not after salvage chemotherapy.

9 · High-risk PC-RPLND

- **Salvage**: viable GCT is at least doubled; OS 60–75%.
- **Desperation**: OS ≈54%. Operate if disease is confined and completely resectable, with markers declining or plateaued, or chemotherapy exhausted. Do **not** operate with rapidly rising markers or multisite progression. **Postoperative marker normalisation** is the only OS predictor.

- **Redo** is mostly for technical failure (retroaortic/retrocaval tissue), so do **complete great-vessel mobilisation the first time**.
- **Late relapse** (yolk sac, somatic-type malignancy) is treated by **surgery first**.

10 · Unique situations

- **Seminoma residual**: dense desmoplasia, viable in ≈10%. If **<3 cm**, observe; if **>3 cm**, do FDG-PET (≥6 weeks †). If avid, choose selected PC-RPLND, salvage chemotherapy or HDCT.
- **Sex cord stromal**: malignant features include >4 cm, necrosis, mitoses and lymphovascular invasion; RPLND is of unproven benefit.

11 · Outcomes, fertility and complications

- **Ejaculation**: template or nerve sparing preserves antegrade ejaculation in **>90%** (primary) at no oncological cost. † If lost, options include sympathomimetics and sperm retrieval.
- Complications: primary 10.6–24%, PC up to 56%. Mortality: primary ≈0, PC <1%.

Complication	Key facts / management
Pulmonary (bleomycin)	PC 3–5%, acute respiratory distress syndrome. Low FiO_2, restrictive fluids; accept SpO_2 **88–90%**
Ileus	Up to 26% (commonest); extraperitoneal approach, no NG tube, early feeding
Chylous ascites	Primary 0.2–2.1%, PC 2–12%; linked to suprahilar dissection (cisterna chyli L1–L2)
Lymphocele	Symptomatic ≤1.7%; drain if infected
Venous thromboembolism (VTE)	Primary <1%, PC 0.1–3.1%; drug prophylaxis if prior chemotherapy, VTE or obesity
Neurological	Femoral neuropraxia; paraplegia with bulky disease

P1 Lymphocele and postoperative lymphatic leak (chylous ascites) — milky tap, triglyceride 2–8× serum, protein >3 g/dL; stepwise management

ALGORITHM

1. Paracentesis; drain if it reaccumulates.
2. Very-low-fat, high-protein diet with medium-chain triglycerides (MCT) and octreotide.
3. If no response, nil by mouth with total parenteral nutrition (TPN).
4. If refractory, lymphangiography with or without embolisation, peritoneovenous shunt or exploration.
5. Continue a low-fat diet ≥ **4 weeks** after resolution.

12 · Landmark trials

Trial	Design	Result	Take-home
Williams (Testicular Intergroup)	RCT, pathological stage II: 2 adjuvant cycles vs observation	Adjuvant near-abolished relapse; equal survival	Observe pN1; individualise pN2
SEMPET	FDG-PET in post-chemotherapy seminoma residuals	>3 cm: specificity 100%, NPV 96%; ≤3 cm PPV 37%	PET for >3 cm only

81 Minimally Invasive Retroperitoneal Lymph Node Dissection

P3 P4 Minimal invasive retroperitoneal lymph node dissection; Robotic RPLND — indications, templates, nerve sparing, robotic steps, lap vs robot, outcomes, AUA/EAU positions, miR-371a-3p

Not in the paper answer:

- The benchmark recurrence-free rate is ≈**85%** for limited pN+ without adjuvant chemotherapy; low MI recurrence is **confounded** by high pN0 and adjuvant rates.

1 · Indications and rationale

- Goals in early-stage disease are to preserve cure, avoid **dual therapy** (surgery and chemotherapy) and minimise lifelong toxicity.
- **Primary** RPLND (retroperitoneal lymph node dissection) is for marker-negative clinical stage (CS) I non-seminomatous germ cell tumour (NSGCT), and small-volume stage II NSGCT and seminoma. It spares chemotherapy in the **20–30%** with enlarged nodes who prove pN0.
- **Post-chemotherapy (PC-RPLND)** is for NSGCT residual **>1 cm** and select seminomas.
- The robot adds fourth-arm retraction and **circumferential great-vessel dissection with lumbar control**.

> **KEY POINT**
> Much recovery gain came from Enhanced Recovery After Surgery (ERAS) pathways, which help every approach, so the benefit of access alone is smaller than it looks.

2 · Guidelines and controversy

- The AUA (early stage) says it **may be offered** by a surgeon expert in germ cell tumours and minimally invasive surgery (MIS), with the patient told long-term data are lacking. The EAU restricts it to **specialist centres** expert in open and MIS.
- With no noninferiority evidence, it is highly controversial. Concerns are staging-only dissection without lumbar division, excess adjuvant chemotherapy, operating on the wrong patients, conversion in **≈5%**, and short follow-up with reported peritoneal carcinomatosis.

3 · Patient selection

- There are no absolute contraindications; relative ones are extensive prior surgery and morbid obesity.
- **Suited to MIS**: chemo-naive high-risk IB; small-volume II (1–2 landing-zone nodes, none **>3 cm**); a post-chemotherapy solitary landing-zone mass with little desmoplasia.
- **Favour open**: most PC-RPLND, a large mass, large chemotherapy shrinkage, possible vessel reconstruction, somatic malignancy or retrocrural disease.

4 · Unchanging operative principles

- Imaging within **≈4 weeks** and markers within **1–2 weeks**.
- **MI-RPLND must completely duplicate open RPLND**: template, completeness and nerve-sparing are unchanged.

ALGORITHM

1. Safe entry and exposure.
2. **Split-and-roll** nodal removal.
3. Prospective nerve sparing.
4. **Lumbar vessel division**.
5. Spermatic cord removal, then closure.

- Chemo-naive patients may have an extended unilateral nerve-sparing template, converted to bilateral if enlarged nodes are found. Post-chemotherapy, the **bilateral full template** is standard.
- Incomplete RPLND or mass-only excision worsens survival.

5 · Preparation, positioning and ports

- Plan for haemorrhage with an **open set in theatre**; after bleomycin, manage fluids and oxygen carefully. A high-fat meal the day before highlights lymphatics.
- Positioning is lateral decubitus, laparoscopic supine, or robotic steep Trendelenburg with a **peritoneal hammock** that allows any template.

6 · Operative steps

- **Right**: incise the Toldt line to the foramen of Winslow and **Kocherise**; divide the gonadal vein at the inferior vena cava (IVC); split over the IVC; clip lymphatics; roll the paracaval packet, spare fibres and divide lumbars. **Do not ligate the right renal artery** in the interaortocaval space.
- **Left**: divide the gonadal vein at the left renal vein and control the **lumbar vein entering the left renal vein**.
- Lower the pneumoperitoneum to check haemostasis; **no routine drain**; discharge at **1–2 days**.

7 · Complications and management

ALGORITHM

1. Major vascular injury: apply pressure and raise the pneumoperitoneum.
2. Inform anaesthesia, get blood and call vascular help.
3. If uncontrolled, keep pressure, undock and convert.

- For MIS repair, **identify the injury before acting: no blind clipping**.
- Prevent chylous ascites with liberal clips rather than energy devices; rates match open, and it is treated by low-fat diet, paracentesis, octreotide and parenteral nutrition.
- Antegrade ejaculation is kept in **≥90%** after primary nerve-sparing.

8 · Outcomes vs open

- There is **no randomised trial**. MIS gives shorter stay and less blood loss but longer operative time; brisk bleeding is the unique risk.
- Robotic node yields approach the open benchmark of **≈30**, but adjuvant chemotherapy is still too frequent. Post-chemotherapy data are too small.

9 · Landmark trials

Trial	Design	Result	Take-home
GETUG T93BP	Good-risk NSGCT: BEP×3 vs EP×4	Most relapses/deaths without or with incomplete RPLND	Complete RPLND matters

82 Tumours of the Penis

1 · Overview

- Squamous cell carcinoma (SCC) is **~95%** of penile malignancy, and **0.4–0.6%** of male cancers in North America and Europe, but up to **10%** in parts of Asia, Africa and South America.
- Prognosis rests on **primary stage and histology** and on **nodal metastasis**.

2 · Premalignant lesions: penile intraepithelial neoplasia (PeIN)

P1 Premalignant conditions in urology (including premalignant lesions of penis) — PeIN definition, WHO 2022 all high grade, HPV-associated vs differentiated, E6/E7/p16, EQ/Bowen/Bowenoid table, LS 2.3–9%, Buschke–Löwenstein, organ-preserving treatment

Not in the paper answer:

- Erythroplasia of Queyrat (EQ) is a red velvety glans lesion in uncircumcised men that mimics balanitis; it progresses in **10–33%**.
- Bowen disease is a scaly shaft plaque mimicking psoriasis or eczema; it progresses in **~5%**. Bowenoid papulosis (HPV-linked shaft papules in young men) **does not progress**.

3 · Aetiology and risk factors of SCC

Factor	Key point
No neonatal circumcision	Neonatal circumcision virtually eliminates invasive cancer; it protects against invasive SCC, **not CIS** (carcinoma in situ). Adult circumcision gives little or no protection
Phimosis	Present in **25–75%**; **7–10×** risk. It mediates the non-circumcision risk, as smegma causes inflammation and fibrosis
HPV	Low risk 6, 11; high risk **16, 18**, 31, 33, 45, 52, 58; **HPV 16 commonest**. Present in **~50%**; wives have ~3× cervical cancer risk
Human immunodeficiency virus (HIV)	~4× risk of penile cancer death; PeIN progresses ~6 years earlier
Tobacco	All forms; dose-dependent and independent of phimosis
Lichen sclerosus (LS)	Cancer in **2.3–9%**; HPV-**independent** SCC
Trauma, obesity	Scarred shaft; obesity causes buried penis and functional phimosis
†**Psoralen and ultraviolet A (PUVA)**	† Genital exposure markedly increases SCC, so shield the genitalia

- **Prevention** is by neonatal circumcision (AAP 2012: benefits outweigh risks), hygiene, no tobacco and HPV vaccination (nonavalent adds 31/33/45/52/58).

> **KEY POINT**
> Two routes converge on the preputial sac: HPV-driven (E6/E7, p16-positive, basaloid/warty) and inflammation-driven HPV-independent (phimosis, smegma, LS). Neonatal circumcision, hygiene and vaccination each prevent part; adult circumcision prevents almost none.

4 · Natural history and spread

- Most arise on the glans or prepuce; **flat ulcerative tumours metastasise earlier**.
- **Buck fascia** is a temporary barrier; once it and the tunica are breached, tumour enters the corpora and spreads by blood.
- Distant metastases are uncommon (**1–10%**) and late. Untreated, death comes within **2 years** from femoral erosion haemorrhage or sepsis.

P1 Lymphatic drainage of testis and scrotum, RPLND templates and penile cancer implications — penile superficial/deep lymphatics, stepwise bilateral spread, no skips

Not in the paper answer:

- Pelvic stations are external iliac, internal iliac and obturator; distant sites (lung, liver, bone, brain) are rare without nodal disease.

5 · Presentation and diagnosis

- It presents as induration, a warty growth or an ulcer, often hidden by phimosis. Examine the lesion and **palpate both groins**.
- **Biopsy is mandatory before definitive therapy**, because depth and grade predict occult nodes and superficial biopsies mislead.
- **Hypercalcaemia without bone metastases** comes from PTH-related substances, tracks bulky inguinal disease and may resolve after node excision. Treat with saline, then bisphosphonate.

6 · Histology

Subtype (WHO 2022)		HPV	Features	Nodal risk
Usual (keratinising)	**45–65%**	Independent	Usually grade II; lymphovascular (LVI) or perineural invasion (PNI) in ~⅓	25–40%
Warty		Associated	Cauliflower, fibrovascular cores; slow	17–25%
Basaloid		Associated	Small cells, necrosis, deep	**>½ node positive**
Papillary		Independent	Low grade, no koilocytosis	≤12%
Verrucous		Independent	**Broad pushing border**	**No metastasis**; recurs if under-excised
Sarcomatoid		Independent	Spindle cells	**Most aggressive**
Rare: pseudoglandular, adenosquamous; clear cell, lymphoepithelioma-like (HPV)		—	Deep invasion	Pseudoglandular, clear cell worst

- **Any anaplastic cells make it grade III**. **LVI** predicts nodes in every study, and **PNI** does so strongly.
- **High-risk HPV or p16 positivity means better disease-specific survival (DSS)**, not better nodal status.

7 · Imaging

A negative scan never clears the groin.

- **Primary**: examination is reliable; **MRI** (with or without artificial erection) helps if equivocal and organ sparing is planned.
- **cN0 groin**: **no imaging detects micrometastases**, so stage surgically; use CT only if obese or after prior groin surgery.
- **cN+ or T3–T4**: CT chest, abdomen and pelvis. **FDG-PET/CT** is best for pelvic and distant disease.

8 · Staging: TNM 8th edition (2017)

T1a versus T1b decides who needs groin staging.

T	Definition
Tis	PeIN
Ta	Non-invasive localised SCC (incl. pure verrucous without destructive invasion)
T1	Glans: lamina propria · Foreskin: dermis/lamina propria/dartos · Shaft: tissue between epidermis and corpora
T1a	**No LVI, no PNI, not high grade**
T1b	**LVI and/or PNI, or high grade (grade III/sarcomatoid)**
T2	**Corpus spongiosum**, with or without urethra
T3	**Corpora cavernosa** (incl. tunica albuginea), with or without urethra
T4	Scrotum, prostate, pubic bone

N	Clinical (cN)	Pathological (pN)
1	Palpable mobile unilateral node	**≤2 unilateral**, no extranodal extension (ENE)
2	Mobile ≥2 unilateral or bilateral	**≥3** unilateral or bilateral
3	Fixed mass or pelvic nodes	**ENE or pelvic nodes**

Stage	TNM
I	T1a N0
IIA	T1b, T2 N0
IIB	T3 N0
IIIA / IIIB	T1–3 N1 / N2
IV	T4, any N3, any M1

- In the 8th edition, **urethral invasion no longer upstages**.

> **KEY POINT**
> Traps: T1b and T2 share stage IIA; any N3 or T4 is stage IV even without metastases; cN counts palpable nodes, pN counts metastases (pN1 allows two). The primary is staged by tissue, the groin by surgery.

9 · Management of the primary tumour

- **Penectomy** is the oncological gold standard (local recurrence **0–8%**), but over half of patients are ≤60, so preserve where safe.
- **Organ preservation** suits PeIN, Ta, T1 and grade I–II in **compliant** men. It is **contraindicated** by size **≥4 cm**, grade III, deep cavernosal invasion, or unfitness for salvage.
- Local recurrence is higher, but **early recurrence is salvaged without survival cost**, except in pT3 or high grade.
- **Margins**: microscopic spread is **~5 mm** for grade I–II and **~10 mm** for grade III. Only margins **<1 mm** raised recurrence, so a **negative frozen-section margin (>1 mm) is enough**; the 2-cm dogma is abandoned.

Stage / site	Treatment
PeIN, glans	Laser, resurfacing; topical alternative
Ta / PeIN — foreskin, shaft skin	Excision to negative margin; laser; topical (Tis only)
Ta, T1 glans	Excision, resurfacing, glansectomy, radiotherapy (RT; not Ta)
T2 glans, no gross cavernosal involvement	**Total glansectomy** with or without distal corporal transection; partial penectomy; RT
T2 with corporal invasion, T3	**Partial or total penectomy**
T4	Neoadjuvant chemotherapy, then surgery in responders
Recurrence after conservation	Re-excision or penectomy; superficial low-grade re-treated conservatively

- **Glans resurfacing** strips the epithelium and adds a split-skin graft, keeping erections.
- **Laser**: CO_2 reaches only ~0.1 mm (recurrence up to 50%), so **Nd:YAG** (up to 6 mm) is preferred, with base biopsies; recurrence is <10% for CIS and 10–25% for T1.
- † Partial penectomy leaves a stump for standing micturition; total penectomy needs a **perineal urethrostomy**.

> **KEY POINT**
> Margin width depends on grade, not a fixed 2 cm. Low-grade tumours spread millimetres and their recurrences are salvageable, so preservation is safe; in high-grade or cavernosal disease local relapse is lethal and amputation regains its place.

10 · Inguinal nodes (see Ch 84)

P2 P3 Importance and management of Inguinal lymph nodes in CA penis; Prognostic index score in carcinoma penis — nodes = strongest prognostic factor, Kroon early vs delayed, risk groups, DSNB, FNAC without antibiotic trial, PLND >2 nodes/ENE, neoadjuvant TIP, ILND morbidity

Not in the paper answer:

- Cure is ~**80%** with ≤2 unilateral nodes and no ENE or pelvic nodes; with positive pelvic nodes, 5-year survival after surgery alone is ~**14%**.
- Occult node rates are **0–18%** for PeIN/Ta/T1a, one-third to over half for T1b and above, and ~60% with corporal invasion.

ALGORITHM

1. **cN0, low risk**: surveillance. If a node appears, do fine-needle aspiration cytology (FNAC); if positive, inguinal lymph node dissection (ILND) and contralateral staging.
2. **cN0, high risk (T1b+)**: bilateral dynamic sentinel node biopsy (**DSNB**) or superficial/fascia-sparing/minimally invasive ILND. If positive, completion ILND.
3. **cN1–2, mobile, <4 cm**: biopsy, then ILND and contralateral staging. If >2 nodes or ENE, add pelvic lymph node dissection (PLND).
4. **Fixed, ≥4 cm or pelvic**: neoadjuvant TIP (cisplatin-eligible) or chemoradiation, then resect responders.
5. **Adjuvant therapy** if >2 nodes, bilateral nodes, ENE or pelvic nodes.

- DSNB refinements (pre-operative ultrasound with or without FNAC, serial sectioning) cut false negatives from ~18% to ~7%. Morbidity is greatest with radical ILND, then superficial or fascia-sparing, then minimally invasive, and least with DSNB.
- No pelvic skip metastases occur, so PLND is mainly staging.

11 · Radiotherapy

- **Circumcise first** to expose the lesion and prevent oedema and phimosis. Surgical salvage remains curative.
- **External-beam RT (EBRT)**: **≥66 Gy in 2-Gy fractions over ~6.5 weeks** with **bolus**; ~55% are recurrence-free at 5 years.
- **Brachytherapy** (Ir-192) suits **T1–T2 glans tumours <4 cm**: ~60 Gy over ~5 days gives ~79% relapse-free survival and ~85% preservation at 5 years, with better erections than EBRT.
- **Late effects** are **meatal stenosis** (10–45%, at 18–24 months) and ulceration; **1 in 5 local recurrences occur after 5 years**.
- For the groins, give **no prophylactic RT** for cN0. Adjuvant RT for **≥2 nodes or ENE** is ≥50 Gy, with **54 Gy for ENE and 57–60 Gy for gross residual**. Chemoradiation (cisplatin if GFR >50 mL/min) treats inoperable nodes.

12 · Systemic therapy

- No randomised trials are completed. **TIP** (paclitaxel, ifosfamide, cisplatin) is most active; **Bleomycin regimens are discouraged**.
- **Neoadjuvant TIP ×4** for bulky N2–N3 or pelvic nodes gives ~50% response, and ~⅔ of responders stay disease-free, so **response selects for surgery**.
- Adjuvant chemotherapy has no proven survival gain but may be considered for pN3 or pelvic nodes.
- Palliative platinum is first line; checkpoint inhibitors help ~1 in 6 in trials.

13 · Follow-up

Groin relapse after observation is usually fatal.

- After organ preservation, use close follow-up and self-examination.
- Observed groins are checked 2–3-monthly for 2 years, then less often. Only ~⅓ of cN0 men with groin relapse survive 5 years.

14 · Non-squamous penile tumours (~5%)

Tumour	Key features / treatment
BCC	Proximal shaft; almost never metastasises. Treat by excision or Mohs
Melanoma	<1%; glans; spreads by blood and lymph; AJCC cutaneous staging. Bilateral modified ILND if Breslow ≥1 mm, ulceration or Clark IV–V (DSNB reasonable); adjuvant nivolumab for stage IIIB–IV
Sarcoma	Malignant are **proximal**; vascular commonest; **local recurrence characteristic**. Superficial: wide excision or partial; deep corporal: total. No ILND unless palpable
Extramammary Paget	Penoscrotal eczematoid plaque; CK7+, CEA+, S-100−; ~½ have another cancer, so screen. Wide excision (up to 3 cm) guided by **mapping biopsies**
Lymphoma	Usually systemic: R-CHOP even if isolated
Metastases	From **bladder, prostate, rectum**; **priapism** is the commonest sign; most die within a year; palliate

15 · Landmark trials

Trial	Design	Result	Take-home
Neoadjuvant TIP (MD Anderson)	Phase II; TIP ×4, cN2–N3 M0	Response 50%; median OS 17.1 months; durable mainly in responders	Standard before surgery for bulky/pelvic nodes
Netherlands Cancer Institute early vs delayed	Early DSNB vs dissection after surveillance relapse	3-y survival 84% vs 35%; ENE near-universal if delayed	Stage high-risk cN0 groins early

84 Inguinal Node Dissection

1 · Why the groin matters in penile cancer

Groin surgery alone can cure node-positive disease.

- Penile squamous cell carcinoma (SCC) has a **locoregional phase**, so the groin is a **curable station**.
- **Nodal status is the strongest prognostic factor**. Extranodal extension (ENE), even in one node, makes it pN3.
- Early versus delayed inguinal lymph node dissection (ILND) for impalpable nodes gave ENE ≈20% vs ≈95% and 3-year disease-specific survival ≈84% vs ≈35%. So manage the groins within **3 months** of diagnosis.
- An adequate dissection removes ≥**8 nodes per side**.

2 · Indications and risk groups

	Low risk	Intermediate	High risk
NCCN	pTis, pTa, pT1a	pT1b G1–2	≥pT2, or G3/G4
EAU-ASCO	pTa, pTis, G1 pT1a	G2 pT1a	≥pT1b
cN0 — NCCN	Surveillance	Modified/radical bilateral ILND or DSNB	Same
cN0 — EAU-ASCO	Surveillance	Surveillance (compliant) vs DSNB	DSNB; ILND if unavailable

NCCN = National Comprehensive Cancer Network; EAU-ASCO = European Association of Urology–American Society of Clinical Oncology; DSNB = dynamic sentinel node biopsy; G = grade. **Trap**: pT1b G2 is intermediate (NCCN) but high (EAU-ASCO).

- For the impalpable high-risk groin, **NCCN leads with ILND** and **EAU-ASCO leads with DSNB**. Select on **primary pathology only**, not HPV or nomograms.
- **cN+**: about half of palpable nodes are metastatic. Do fine-needle aspiration cytology (FNAC), then excisional biopsy if negative. If any is positive, do **bilateral radical ILND**. Give no antibiotic trial.
- With unilateral palpable nodes, the other side has micrometastasis in ≈**30%**, so stage it.
- **Bulky or fixed (cN3)**: give cisplatin–taxane chemotherapy first, then surgery. Resect upfront only if chemotherapy-ineligible or vessel invasion is imminent; involved skin needs a myocutaneous flap.

3 · Staging work-up

- Palpable nodes or high risk need 18FDG-PET/CT or CT chest–abdomen–pelvis. Imaging misses deposits <1 cm, so it has no role in the impalpable groin.
- **FNAC** is specific but insensitive: if positive, do ILND; if negative, do DSNB or ILND and never discharge.
- **DSNB**: do ultrasound, with or without FNAC, first, because a **tumour-replaced node** reroutes tracer and gives a false negative. Then use ^{99m}Tc nanocolloid and blue dye with serial sectioning. Positive: ILND †; negative: surveillance.

KEY POINT

Every groin test fails toward **false negatives**; a negative result never ends the pathway.

4 · Lymphatic anatomy

P2 Importance and management of Inguinal lymph nodes in CA penis. — superficial/deep systems, bilateral stepwise no-skip drainage, Daseler's five zones (superomedial = sentinel), superficial 4–25 and deep 1–3 nodes, Cloquet

Not in the paper answer:

- The **femoral triangle** is bounded by the inguinal ligament, **adductor longus** medially and **sartorius** laterally.
- Raise flaps deep to **Camper's fascia**, which holds the capillary network; too thin causes necrosis and too thick leaves nodes.
- In the fossa ovalis the order, lateral to medial, is **nerve, artery, vein**, so **never dissect lateral to the femoral artery**.

5 · Templates

Template	Extent	Features
Radical	Whole femoral triangle, superficial and deep	Fascia lata opened; saphenous ligated; sartorius transposed
Modified (Catalona)	Excludes lateral to femoral artery and caudal to fossa ovalis	Shorter incision, saphenous preserved, thicker flaps, no transposition
Superficial	Above fascia lata only	Fascia lata preserved

Modified ILND nodes go to frozen section; if **positive, convert to radical** †.

6 · Open radical ILND

ALGORITHM

1. Frog-leg position; incision ≈2 cm below and parallel to the inguinal ligament.
2. Raise flaps 3 cm above the ligament to the apex; divide the saphenous at the apex (spare if disease minimal).
3. Open fascia lata over sartorius and adductor longus; dissect cephalad on the femoral vessels.
4. Divide the saphenous at the junction; clear deep nodes up to the femoral canal.
5. **Sartorius transposition**; close the femoral canal.
6. Suction drains; tack the flaps.

7 · Pelvic lymph node dissection (PLND)

- If the inguinal frozen section is positive, do PLND at the same sitting; otherwise stage it.
- † EAU: with ≥2 positive inguinal nodes or ENE (pN2–3), do ipsilateral PLND, open or minimally invasive.

8 · Video-endoscopic ILND (VEIL)

P4 Videoendoscopic inguinal lymphadenectomy — same template beneath Scarpa's, 15 mm Hg, frozen section gates deep dissection; similar yield, fewer wound complications; non-bulky nodes only

- The commonest robotic complication is lymphocele.

9 · Complications and post-op care

Fear of complications must not cause omission.

Complication	Cause	Prevention / management
Wound infection, breakdown (commonest)	Skin flora, dead space	Asepsis, tack flaps, incisional negative pressure
Flap necrosis	Breach of Camper's capillary network	Correct plane, modified template, VEIL; debride, then flap or negative pressure †
Lymphocele / seroma	Unsealed lymphatics	Sealing devices; drain until **<30 mL/day**; drain symptomatic or infected collections †
Lymphoedema	Lymphatic interruption	Saphenous sparing, modified template, compression hose ≥1 month
Others	Phlebitis, neurapraxia, pulmonary embolism (PE), femoral bleeding	Thromboprophylaxis; stay medial to artery

Mobilise early. NCCN advises antibiotic prophylaxis, though the authors do not use it routinely.

10 · Follow-up

Most groin relapses occur within 2–3 years (median ≈1 year). EAU-ASCO advises examination 3-monthly for 2 years, then 6-monthly to 5 years, with imaging only if node-positive; NCCN images everyone.

83 Tumours of the Urethra

1 · Benign urethral tumours

Lesion	Who / where	Clue	Treatment
Leiomyoma	Reproductive-age women; posterior wall	Firm suburothelial mass; grows in pregnancy	Excision; may recur
Haemangioma	Men, any age	Bluish sessile lesion, painless bleeding	Fulguration/laser; open excision if extensive
Fibroepithelial polyp	Boys; bulbar/posterior	Polyp on a stalk; filling defect	Transurethral resection

2 · Anatomy, histology and lymphatics

Tumour site predicts its type and its nodal basin.

Feature	Male (~20 cm)	Female (~4 cm)
Segments	Anterior: fossa navicularis, pendulous, bulbar. Posterior: membranous, prostatic	Anterior = distal ⅓; posterior = proximal ⅔
Epithelium	Urothelium in the prostatic urethra, columnar in the anterior urethra, squamous in the fossa navicularis	Proximal ⅓ urothelium; distal ⅔ stratified squamous
Histology	Urothelial carcinoma (UC) 50–80%, squamous cell carcinoma (SCC) 10–30%, adenocarcinoma 5–10%. **Anterior: up to 80% SCC; posterior: mostly UC**	UC 28–45%, SCC 19–29%, **adenocarcinoma 28–38%** (near-equal)
Lymphatics (both sexes)	Anterior drains to **inguinal** nodes (bulbar also external iliac); posterior drains to **pelvic** nodes	

KEY POINT

Site predicts histology (distal SCC, proximal UC), histology predicts behaviour, and the same anterior–posterior line fixes the nodal basin. In women it is also the **continence boundary**: the distal third can be excised without incontinence.

3 · Staging: AJCC 8th edition

T	Penile urethra and female urethra	Prostatic urethra
Ta / Tis	Non-invasive papillary / carcinoma in situ (CIS)	Tis also includes prostatic duct involvement **without stromal invasion**
T1	Subepithelial connective tissue	Subepithelial connective tissue
T2	Corpus spongiosum or periurethral muscle	Prostatic stroma
T3	Corpus cavernosum or anterior vagina	Periprostatic fat
T4	Other adjacent organs (bladder)	Bladder, rectum
N	N1 single, N2 multiple regional nodes (inguinal, pelvic, presacral): **number, not size**	

- **Evaluation** is by examination under anaesthesia with cystourethroscopy, bimanual palpation and biopsy, plus **MRI** for local extent and CT chest if invasive.

- **Palpable inguinal nodes (20–30%) are almost always metastatic**, so, unlike penile cancer, no antibiotic trial is given.

4 · Male primary urethral cancer (PUC)

Distal tumours are curable by surgery; bulbar tumours need multimodal therapy.

- Patients are >60 years. **Urethral stricture** is the commonest risk factor (up to 50%); others are lichen sclerosus, pelvic radiotherapy (RT) and HPV 16.
- † Biopsy a stricture that bleeds or recurs rapidly.
- **Prognosis** depends on stage and location: distal tumours do 2–3-fold better, and bulbar survival is as low as 20–30%.

Pendulous / fossa navicularis (distal)

- Surgery alone controls low-stage disease, ranging from transurethral resection to **distal urethrectomy with or without partial penectomy**, to total urethrectomy with perineal urethrostomy, with or without total penectomy.
- **Penile preservation** is endorsed by EAU and NCCN, using options such as topical 5-fluorouracil (5-FU) for distal CIS, distal urethrectomy, or glansectomy. A positive margin needs re-excision or adjuvant RT.
- A **margin of 5 mm** is now standard (1 cm traditionally).
- **Primary RT** is only for early anterior lesions in men refusing surgery, and it is ineffective for bulbar PUC.

Bulbar (proximal anterior)

- The few low-stage tumours are treated by transurethral resection, or segmental excision and anastomosis.
- **Advanced** disease needs radical cystoprostatectomy, pelvic lymphadenectomy, total urethrectomy and penectomy. Surgery alone gives ~¼ 5-year survival, so use **multimodal therapy**.

Inguinal nodes (both sexes)

- There is **no proven benefit of prophylactic inguinal dissection in cN0**. For cN1, give neoadjuvant chemotherapy or chemoradiotherapy, then lymphadenectomy.

5 · Male posterior (prostatic) urethral cancer

Only stromal invasion needs cystectomy.

- It is usually **secondary to bladder UC**, found in 20–40% at radical cystectomy (RC), so always evaluate the bladder and upper tracts.
- Diagnose by TURP biopsy; it is often CIS. BCG for high-grade superficial disease gives ≥70% response.

Involvement	Treatment	5-year survival
Mucosal	TURP + BCG	67–74%
Ductal / acinar (still Tis)	TURP + BCG, or RC with or without urethrectomy	67–74%
Stromal invasion	**RC + urethrectomy + pelvic lymphadenectomy**, with or without neoadjuvant chemotherapy	~36%; follows bladder stage

KEY POINT

The only question is "stroma breached?" TURP before BCG debulks, improves contact and stages. Ductal disease is not T2, so do not jump to cystectomy.

6 · Female urethral cancer

- It is ~⅓ of PUC, in women >60 years. Risks include chronic irritation, HPV and **urethral diverticulum (6–10% harbour malignancy**, mostly adenocarcinoma).
- Distal tumours are SCC; proximal ones are UC or adenocarcinoma, which is **mucinous** or **clear cell** (in diverticula).
- **Location decides continence-sparing excision**; stage with MRI pelvis.
- **Stage dominates prognosis**: distal tumours do better than proximal, which do better than pan-urethral.

Treatment

Situation	Management
Small non-invasive	Endoscopic resection or laser, with or without BCG
Low-stage distal	**Distal urethrectomy** (transvaginal) **or RT** (brachytherapy with or without external beam, ~60 Gy): equal, ≥70% cured. Stress incontinence (>40%), meatal stenosis; ~1 in 5 recur locally. RT complications 20–50% (stricture, necrosis, fistula)
Proximal / pan-urethral	**Anterior exenteration**: RC, total urethrectomy, extended pelvic lymphadenectomy and anterior vaginal wall, with or without hysterectomy and pubic rami. Alone, most die, so use **multimodal** therapy
Palpable groin nodes	Chemoradiotherapy, chemotherapy or lymphadenectomy; often curable. Pelvic nodes are poor

- RT local control depends on the **length of urethra involved**, not histology or size.

7 · Multimodal therapy (both sexes)

- For advanced or proximal disease, give **cisplatin-based neoadjuvant chemotherapy** (EAU, NCCN) or **chemoradiotherapy**, then consolidative surgery. Response predicts survival.
- **Chemoradiotherapy** is 45–55 Gy to the genitalia, perineum and groins with 5-FU and mitomycin C. It is preferred for advanced SCC, though every responder develops a stricture.

Histology	Regimen
UC	MVAC (methotrexate, vinblastine, doxorubicin, cisplatin); gemcitabine–cisplatin
UC or SCC	Paclitaxel, ifosfamide, cisplatin (TIP); cisplatin, gemcitabine, ifosfamide
Adenocarcinoma	Gemcitabine, 5-FU, leucovorin, cisplatin

Stage (NCCN, anterior male / female)	First line
Tis, Ta, T1	Repeat transurethral resection, with or without intraurethral BCG or chemotherapy
Male T2 pendulous	Partial urethrectomy with or without partial penectomy; if margin positive, chemoradiotherapy, surgery or RT

Stage (NCCN, anterior male / female)	First line
Male T2 bulbar	Urethrectomy with or without RC; if ≥pT3/N1, adjuvant chemotherapy or chemoradiotherapy
Female T2	RC + urethrectomy, or distal urethrectomy by location, or chemoradiotherapy
≥T3 N0	Chemoradiotherapy with or without surgery; or neoadjuvant chemotherapy, then surgery or RT
N1	Chemoradiotherapy (preferred for SCC) or systemic therapy, with or without consolidative surgery

- Recurrence after resection of invasive disease is ~50%, so survey the urethra, groins and body.

8 · Urethral recurrence (UR) after radical cystectomy

Urethral risk decides whether a neobladder is safe.

Male

- Risk is **1–15%** (~5%), mostly within 2 years.
- **Risk factors** are **prostatic urethral involvement** (15% vs 5%), **positive urethral margin**, cutaneous diversion, CIS and multifocality. An orthotopic neobladder (ONB) is protective (~5% vs 9%).
- Routine prophylactic urethrectomy is abandoned, because 64–100% of specimens are negative.
- **Before ONB**, confirm a negative intraoperative frozen section of the urethral margin.
- **Urethrectomy** is absolute for a positive margin or gross urethral tumour, and relative for diffuse CIS, prostatic urethral involvement or multifocal disease. Delayed urethrectomy is as good as simultaneous.

Female

- The main risk is **bladder neck involvement**, but only 30–50% of these have urethral tumour, so it is not an absolute contraindication.
- ONB is allowed if the bladder neck, urethra and anterior vaginal wall are uninvolved and frozen section is negative; UR after ONB is <5%.

Surveillance and treatment

- Urethral wash cytology has no proven survival benefit; if positive, do urethroscopy and biopsy.
- **Total urethrectomy is treatment of choice**, including the meatus; after ONB, convert to cutaneous diversion. **Intraurethral BCG treats CIS** (not papillary or invasive). Invasive UR has median survival ~17 months.

85 Benign Disorders of the Penis and Urethra

1 · Reconstructive surgical technique

- **Cautery**: use **bipolar on the urethra** to limit spongiosal damage, and monopolar on subcutaneous tissue and muscle.
- A **bougie à boule** calibrates. **Goodwin sounds** dilate over a guidewire, so they are safer than van Buren sounds.
- **Sutures** are always **absorbable**, because permanent ones form a stone nidus; monofilament polydioxanone (PDS) is preferred. Support is needed for **30–60 days**, so avoid chromic, which lasts 14 days. Use **5-0** for most urethral work and **6-0** for a single-layer mucosal anastomosis (non-transecting excision and primary anastomosis, EPA).

2 · Tissue transfer: grafts and flaps

P1 Enlist various grafts used in urology and role of BMG in urology — graft vs flap; take by imbibition then inosculation (~48 h each), immobilise 5–10 days; STSG vs FTSG; other grafts; BMG properties, buccal = lingual

Not in the paper answer:

- **Extragenital full-thickness skin graft (FTSG)** fails in the urethra because its greater mass makes take ischaemic. The exception is the **posterior auricular (Wolfe) graft**, which behaves like a split-thickness skin graft (STSG).
- Oral mucosa graft (OMG) has a **panlaminar plexus** and sacculates less than skin. Buccal onlay failures occur in the **first 12 months**.

> **KEY POINT**
> Graft trade-off: a dense fine plexus and low mass favour take, while carried dermis or lamina propria gives durability and resists contraction. This explains STSG vs FTSG, the failure of extragenital FTSG, the Wolfe exception, and why OMG is ideal.

Flaps

- Flaps are classified by **composition** (fasciocutaneous, fascial, musculocutaneous, muscle or visceral), **blood supply** (axial on a named vessel, or random as in a Z-plasty), **movement** (advancement, rotation or transposition) and **location** (local, distant pedicled or free).

3 · Surgical anatomy

Fascial layers decide flap planes and how blood spreads.

- Layers, deep to superficial, are **tunica albuginea, Buck's fascia, dartos, skin**. Dartos is fat-free, carries the skin vessels and fuses with Colles fascia.
- **Skin supply**: the superficial external pudendal artery feeds a subdermal plexus, which is the basis of the McAninch circular flap.
- **Deep supply**: the internal pudendal artery continues as the common penile artery, giving the **bulbourethral, dorsal and cavernosal** arteries.

Situation	Spread
Buck's intact (penile fracture)	Confined to the shaft, giving an **eggplant deformity**

Situation	Spread
Buck's ruptured (straddle)	Within Colles–Scarpa: scrotum, perineum, abdominal wall; limited on the thigh by fascia lata
Fournier gangrene	From perineum and scrotum up the abdominal wall via Scarpa

KEY POINT

Two envelopes, two vascular worlds: dartos carries the skin supply (so skin flaps are raised on dartos); Buck's encloses the erectile bodies fed by the common penile artery.

4 · Lichen sclerosus (LS)

P1 Lichen sclerosus (BXO, balanitis xerotica obliterans) — meatus sentinel, Littre-gland stricture mechanism, spares proximal urethra, biopsy features, RUG first, clobetasol, oral mucosa never skin, perineal urethrostomy

Not in the paper answer:

- Malignant transformation (mostly SCC) reaches up to **8%**.
- **Steroid**: clobetasol propionate or betamethasone 0.05% once or twice daily for 1–2 months, then on alternate days for 3–4 months.
- Meatal stenosis is treated by extended meatotomy or BMG onlay, and early repair may avert a panurethral stricture.

KEY POINT

LS is a progressive skin disease sparing the proximal urethra: hence meatus as sentinel, retrograde urethrography (RUG) before meatotomy, oral mucosa not skin, high recurrence, and easy perineal urethrostomy.

5 · Other urethral lesions

Lesion	Key features	Management
Urethral haemangioma	Bloody discharge; often extends beyond view, so image with **MRI**; benign	Observe; small: laser or triamcinolone; large: excision, staged repair
Reactive arthritis (Reiter)	Arthritis, conjunctivitis, urethritis after Chlamydia/Shigella/Salmonella; HLA-B27 in ~2/3; **circinate balanitis**	Urethritis is self-limiting
Amyloidosis	Urethral mass, obstruction, haematuria; needs biopsy	Observe; dilate or urethroplasty if obstructive
Urethrocutaneous fistula	Mostly iatrogenic (hypospadias, phalloplasty)	Treat the **underlying process** as well as the tract
Male urethral diverticulum	Congenital (all layers) or acquired (no smooth muscle; behind stricture, catheter, trauma); dribbling, UTI, stone	Cystourethroscopy and RUG; repair and treat the cause
Enlarged utricle	Müllerian remnant, with proximal hypospadias	A catheter may enter the utricle; UTI, post-void dribble

6 · Prepuce and meatus

- **Paraphimosis (emergency)**: a trapped foreskin causes oedema and can lead to glans necrosis. Reduce by steady compression, with the thumbs pushing the glans and the fingers pulling the foreskin. If irreducible, do an emergency **dorsal slit**; offer elective circumcision for recurrence.
- **Phimosis** is physiological in infants (resolving at 1–8 years); in adults it leads to balanitis and cancer. Treat with steroid, dorsal slit or circumcision.
- **Balanitis**: **Candida** is commonest. Treat with hygiene and antifungal, and **check blood glucose**.

P2 Management of Urethral meatal stenosis. — adult causes, RUG first, ventral meatotomy with 3 sutures, steroid self-dilation, urethroplasty if fossa involved

Adult circumcision (sleeve)

ALGORITHM

1. Penile block. Foreskin retracted: mark the cuff incision across the frenular base; incise to Buck's.
2. Foreskin advanced: mark the coronal incision.
3. Excise the sleeve using **bipolar**, because adult vessels do not seal by pressure.
4. Quadrant dartos sutures, then skin closure.

- **Human immunodeficiency virus (HIV)**: the foreskin has many CD4+ cells and microtears. Circumcision lowers female-to-male HIV by ~**60%**, and also lowers herpes simplex and HPV.

7 · Penile curvature

P1 Congenital Penile curvature.. — chordee without hypospadias vs congenital curvature (young man, large erection); never divide a urethra reaching the glans tip

Not in the paper answer:

- Congenital curvature is a smooth whole-shaft curve, whereas chordee without hypospadias shows a hooded prepuce and ventral inelasticity.

Devine–Horton type	Abnormal tissue
I	Spongiosum deficient (chordee without hypospadias, I–III)
II	Dysgenetic Buck's and dartos; urethra/spongiosum normal
III	Dartos only
IV	All normal; asymmetric tunica = **congenital curvature**
V	Congenital short urethra; its existence is doubted

ALGORITHM

1. Excise ventral dysgenetic tissue; mobilise spongiosum from glans to perineum.
2. Artificial erection; ventral midline septotomy.
3. If still curved, do dorsal tunical ellipse excision or plication under the elevated neurovascular bundle.

- **Acquired non-Peyronie curvature** (see Ch 76) follows deep urethrotomy, neglected or **subclinical fracture**, or a septal tear. Erections are normal, with no shortening. Treat by **scar excision and graft**, not contralateral plication.

8 · Penile transplantation

A composite allotransplant restores appearance, erection and voiding after penile loss, but needs lifelong immunosuppression; the alternative is phalloplasty.

9 · Adult acquired buried penis (AABP)

It is self-perpetuating and usually needs surgery.

- Risks are **obesity**, aggressive circumcision, scrotal lymphoedema and LS.
- It is a **vicious cycle**: the escutcheon tethers the penis, and trapped urine causes inflammation, LS and scarring, which trap it further. So **weight loss alone rarely cures**.
- The **Pariser** classification grades the repair needed: I skin flap, II graft, III scrotoplasty, IV escutcheonectomy, V panniculectomy.
- **Evaluate the urethra**, because a severe stricture needs urethroplasty **before** grafting.

ALGORITHM

1. Slit the phimotic ring and place glans traction.
2. Circumcising incision; excise non-viable skin.
3. Do escutcheonectomy (harvest the graft from it; keep 2–3 cm of skin at the base).
4. Tack Scarpa to the rectus fascia or pubis; place drains.
5. Cover with STSG (0.012–0.020 inch), fibrin glue and a bolster for 5–7 days.

- **Scrotal lymphoedema** may be congenital, infective (**Wuchereria bancrofti**, lymphogranuloma venereum), malignant or after node dissection or radiation. Resect **all** involved tissue and cover with spared perineal or lateral scrotal skin.
- **Foreign body injection** (silicone, paraffin) causes fibrosis, necrosis and abscess. Excise it all and graft, and screen for **body dysmorphic disorder**.

10 · Landmark trials

Trial	Design	Result	Take-home
African circumcision trials (South Africa, Kenya, Uganda)	Three randomised controlled trials (RCTs), adult male circumcision vs none	~60% reduction in female-to-male HIV acquisition	Circumcision is an HIV-prevention strategy

86 Urethral Stricture Disease

1 · Definitions, anatomy, aetiology

P2 P3 Urethral stricture disease: etiology, presentation and management (including recent advances); Urethral stricture: classification, DVIU and surgery after failed VIU — anterior-only definition, spongiofibrosis, area ∝ diameter², Jordan spectrum, dorsal eccentric bulb, dual supply, aetiology table (idiopathic ≈ iatrogenic), LS via Littre glands

Not in the paper answer:

- Straddle strictures often present late; prostate cancer treatment causes posterior stenosis.

Site	Correct term
Anterior urethra	Stricture (epithelial scar with or without spongiofibrosis)
Membranous urethra after pelvic fracture	Distraction defect; pelvic fracture urethral injury (PFUI)
Posterior urethra after trauma/radical prostatectomy	Posterior urethral stenosis
Prostatic urethra, bladder neck	Contracture or stenosis

- The common penile artery supplies the corpus spongiosum. The bulbar urethra lies dorsally, so a deep 12 o'clock urethrotomy in the bulb is dangerous.
- In lichen sclerosus (LS), microabscesses form in the glands of Littre; antibiotics relieve symptoms, not the stricture.

> **KEY POINT**
> Retrograde glans flow allows the bulb to be transected and mobilised. Where it is lost (hypospadias, common penile artery injury), mobilisation causes ischaemia and PFUI repair failure.

> **KEY POINT**
> Aetiology predicts biology: iatrogenic and straddle strictures are focal scars suited to excision; LS is a progressive skin disease that recurs in any skin used to repair it.

2 · Presentation and evaluation

- **Acute retention**: if a catheter fails, do a retrograde urethrogram (RUG) and place a **suprapubic cystostomy**, not blind dilation. Pass any guidewire under flexible endoscopic vision.
- Four determinants guide treatment: **location, length, depth, density**.
- The treating surgeon does a dynamic RUG and voiding cystourethrogram (VCUG). Do not dilate at first urethroscopy. MRI is for tumours and PFUI only.
- **Hydraulic trap**: voiding pressure holds the urethra above the stricture open, so a segment that cones down is scarred.
- **Urethral rest** with a suprapubic tube for **6–8 weeks** reveals the true length.

3 · Decision making

- First agree on **palliation or cure**. The **reconstructive ladder is archaic**: go to the procedure most likely to cure.

> **ALGORITHM**
>
> 1. If bulbar, <1.5 cm, without dense spongiofibrosis and untreated, then dilate or do one internal urethrotomy; if it recurs, then urethroplasty.
> 2. If bulbar 1–2 cm (up to 3–5 cm proximally), then excision and primary anastomosis (EPA).
> 3. If longer bulbar, then oral mucosa graft (OMG) onlay, with or without augmented anastomosis.
> 4. If penile, then onlay (graft or flap); if hostile bed, LS or failed hypospadias, then staged OMG.
> 5. If panurethral, options exhausted or elderly, then perineal urethrostomy.

4 · Dilation and internal urethrotomy (DVIU)

P3 Urethral stricture: classification, DVIU and surgery after failed VIU — secondary-intention race, 10–2 o'clock thin spongiosum, complications, second urethrotomy futile, self-dilation, mitomycin C, laser = cold knife, catheter 3–7 days

Not in the paper answer:

- DVIU does worse when luminal narrowing exceeds **74%**; the safest dilation is a **balloon** over an endoscopically placed guidewire.

Favourable	Unfavourable
Bulbar	Penile, fossa navicularis
<1.5 cm	Longer
No dense deep spongiofibrosis	Dense (e.g. straddle)
First procedure	Repeat

- Self-catheterisation tapers over **3–6 months**; the stricture recurs when it stops. A 6-week catheter is no better than 3–7 days.

> **KEY POINT**
>
> No endoscopic adjunct removes spongiofibrosis, so endoscopy cures only the short superficial bulbar stricture; each repetition spends the patient's reconstructive capital.

5 · Excision and primary anastomosis (EPA)

- EPA is the **gold standard** for short bulbar strictures; recurrence is rare. Rules: **excise all fibrosis, spatulate widely, keep it tension-free**.
- Length comes from spongiosal mobilisation, freeing Buck's fascia and the intracrural space, and detaching bulbospongiosus from the perineal body. **The more proximal the stricture, the longer the excisable segment.**

	Transecting	Vessel-sparing transecting	Non-transecting
Bulbar arteries	Divided	Preserved	Preserved

	Transecting	Vessel-sparing transecting	Non-transecting
Steps	Excise, spatulate, anastomose	Divide at membranous–bulbar junction, dorsal spatulation	Dorsal stricturotomy, excise scar from within, close transversely
Use	Most	Proximal bulbar	Proximal

Vessel-sparing protects men with poor retrograde supply (hypospadias, trauma, prior surgery); its other benefits are unproven.

6 · Substitution urethroplasty

- **Buccal mucosa graft (BMG)** is the workhorse, one-stage or staged. Skin grafts are not used in LS. Grafts do best in the **bulb**, which has a thick vascular bed.
- Graft position may be ventral (with spongioplasty), lateral, or **dorsal (Barbagli)**, spread-fixed on the corpora.
- **Augmented anastomosis** often beats pure onlay. **Kulkarni** is a one-stage panurethral dorsal onlay via the perineum with **penile invagination**.
- **Staged**: stage 1 lays the urethra open and quilts the graft to dartos. Stage 2 tubularises it to about 28 Fr after at least 6 months. It is used for LS and failed hypospadias.

> **KEY POINT**
> A graft lives on its bed: it succeeds where the bed is thick and vascular (bulb, dorsal corpora), and is staged when the bed, or skin disease, cannot be trusted in one step.

- **Flaps** use hairless genital skin on dartos, such as the **Orandi** ventral longitudinal flap or a **circular island** flap up to **15 cm**; fossa navicularis strictures use a Jordan or De Sy island.
- Grafts equal flaps; flaps suit irradiated or multiply operated urethras. **Onlay beats tube**.
- **LS**: skin repairs fail, so use **oral mucosa only**, often staged.
- **Perineal urethrostomy** opens the urethra dorsally to healthy mucosa, sutures it to skin, and can add a BMG to complete the dorsal wall.

7 · Meatal stenosis

- **Child**: post-circumcision meatitis needs ventral meatotomy and parental dilation; early meatitis responds to 0.05% clobetasol.
- **Adult**: the fossa is usually involved, so evaluate it as a stricture before ventral meatotomy or dorsal Y-V plasty.

8 · Pelvic fracture urethral injury (PFUI)

P3 Describe early and late management of PFUD injuries — bulbomembranous site, primary realignment debate, cystogram + RUG, bladder neck not predictive, MRI, delayed perineal repair after 3 months, progressive steps, voiding study 21–28 days

Not in the paper answer:

- PFUI occurs in about **10%** of pelvic fractures. Repair at **3–6 months**; endoscopic core-through is not curative.
- Failure is mostly **ischaemic stenosis** of the mobilised bulb; internal pudendal status decides the order of surgery (table below).

ALGORITHM

1. On-table endoscopy; exaggerated lithotomy; perineal incision.
2. Mobilise the bulb; pass a sound via the suprapubic tract; excise fibrosis.
3. Spatulate to **≥30 Fr**; place 8–10 tension-free 4-0 PDS sutures over a silicone stent.
4. If tension remains, then add steps in order: mobilisation, intracrural separation, infrapubectomy, corporal rerouting.

- Success is in the high **90s %**.

Internal pudendal status	Potency	Action
Intact ≥1 side	Often potent	Reconstruct
Reconstituted only	Never potent	Reconstruct; revascularise for ED later
Bilateral occlusion	—	**Revascularise first** (ischaemic stenosis risk)

Screen at-risk men with **duplex ultrasound, then angiography if abnormal**.

9 · Vesicourethral stenosis and fistula

- Stenosis after prostate cancer treatment is incised deep at 3 and 9 o'clock, adding mitomycin C if recalcitrant. Obliteration needs robotic or perineal repair, often followed by an artificial urinary sphincter.
- **Rectourethral fistula**: a simple one (<2 cm, non-irradiated) is repaired perineally or by York-Mason. A complex (irradiated) one needs faecal diversion and **gracilis interposition, with or without BMG**.

10 · Complications and follow-up

- Recurrence is early after EPA and within 12 months after grafts. ED is uncommon, higher with long repairs and straddle injuries, and often improves.
- Follow up with symptoms, uroflow and post-void residual, and urethroscopy if suspicious.

11 · Landmark trials

Trial	Design	Result	Take-home
ROBUST III	Phase III RCT, Optilume (paclitaxel balloon) vs endoscopic control after multiple failed endoscopic treatments	1-year stricture-free 74.6% vs 26.8%	Drug-coated balloon beats repeat endoscopy; role as first-line undefined

87 Surgery of the Scrotum and Seminal Vesicles

1 · Surgical anatomy of the scrotum

- Lax, rugated skin gives **poor tamponade**, so haematomas keep enlarging and haemostasis must be meticulous.
- Dartos keeps necrotising fasciitis off the testes. **Transdermal absorption** is about **40×** that of other skin, and the skin is rich in 5α-reductase.

Territory	Artery	Parent
Scrotum, anterior	Superficial and deep external pudendal	Femoral
Scrotum, posterior	Posterior scrotal	Perineal (internal pudendal)
Testis, primary	Testicular	Aorta
Testis, secondary	Deferential	Superior vesical, from internal iliac
Testis, secondary	Cremasteric	Inferior epigastric, from external iliac

KEY POINT

Three trunks (aorta, internal iliac, external iliac) anastomose in the cord and at the cauda. So the testicular artery can be ligated in Fowler–Stephens orchidopexy, and the superior epididymal artery sacrificed in vasectomy reversal.

2 · Peri-operative principles and access

- Shave with a **razor on the day of surgery**. The American Urological Association (AUA) advises a single IV cephalosporin or clindamycin **only with risk factors**.
- Benign disease uses a scrotal incision. **Possible malignancy** needs an inguinal incision, with the cord clamped before the testis is handled.
- **Always close the dartos**, because most haematomas come from inadequate dartos closure.

3 · Scrotal wall: excision, scrotectomy, reconstruction

- **Fournier gangrene** is the commonest indication for partial scrotectomy. Give fluids and broad-spectrum antibiotics, and **debride urgently**, because it spreads **2–3 cm/h**; mortality is about **20%**. Re-debride every **24–48 h**.
- Scrotal violation in non-seminomatous germ cell tumour (NSGCT) calls for partial scrotectomy of the scar.
- If more than **50–60%** of skin is lost, do not reconstruct at once; protect the testes in thigh pockets or under a vacuum dressing (VAC).

ALGORITHM

1. If <50% and tension-free, then close primarily.
2. If <50% with tension, then local scrotal advancement or secondary intention.
3. If >50%, then VAC, then stretched scrotal advancement or delayed primary closure.
4. If no local tissue, then fasciocutaneous thigh flap; meshed split-thickness skin graft (STSG) last.

KEY POINT

The scrotum works because it is lax. Graft contracture defeats that, whereas residual scrotal skin stretches enormously and is the best donor.

4 · Hydrocele

- **Congenital** hydrocele comes from a patent processus; about **80%** resolve by 18 months. **Acquired** hydrocele reflects a production–absorption imbalance.
- **Colour Doppler ultrasound first** excludes tumour and decides the scrotal or inguinal route.

Technique	Steps	Best for
Lord plication	Small opening; radial plicating sutures; no dissection	Thin-walled, smaller; least haematoma, no drain
Jaboulay (Winkelman)	Excise part of parietal layer, evert remainder behind testis	Thin, floppy sacs
Bottleneck	Trim to 2-cm cuff, tack edges	Excisional
Excision	Excise sac, oversew edges	Thick, large, multiloculated; lowest recurrence

KEY POINT

The more tunica excised, the lower the recurrence but the more haematoma. Recurrence after any successful repair is **0–6%**.

- **Aspiration and sclerotherapy** suit the unfit; aspiration alone recurs, so add a sclerosant such as doxycycline. With **retreatment**, resolution is **85–96%**, but results are poor if multiloculated or more than **750 mL**.

5 · Spermatocele and epididymal surgery

- Operate on a spermatocele only for symptoms, and **counsel about obstructive azoospermia**.
- A suspected malignant epididymal tumour is approached inguinally; a proven benign one by scrotal microsurgical excision.
- **Epididymectomy** gives **poor results for chronic epididymitis** (>6 weeks), so it is a last resort; it works better with a **palpable lesion**.

6 · Varicocele repair

P2 Surgical management of bilateral varicocele — ~15% of men; Palomo, laparoscopic, microsurgical subinguinal (lowest recurrence and hydrocele, gold standard) and embolisation compared; microsurgical steps

Not in the paper answer:

- †Embolisation suits recurrence after surgery. For orchialgia, **microsurgical repair** gives more complete relief.

7 · Vasectomy

- **Counsel**: the man is not sterile at once, so contraception continues until post-vasectomy semen analysis (PVSA). Repeat vasectomy is needed in <**1%**; pregnancy occurs in about **1 in 2000** even after an azoospermic PVSA; chronic pain affects about **1–2%**.

- Give **no routine antibiotics**. **Isolation** determines morbidity; **occlusion** determines efficacy.

Feature	Conventional (CV)	Minimally invasive (MIV)	No-scalpel (NSV)
Skin opening	1.5–3 cm	≤ **10 mm**	Puncture <10 mm
Status	Difficult cases	**Recommended**: less pain, bleeding, infection	Original MIV (Li Shunqiang, China, 1974)

- Open on the **straight vas**, because the convoluted vas is harder now and at reversal.

AUA-recommended occlusion (failure ≤1%)	Note
Mucosal cautery (MC) and fascial interposition (FI)	FI = internal spermatic fascia over one end
MC alone (no clips/ligatures)	Cautery of lumen 0.5–1.5 cm
Open-ended: abdominal end MC and FI, testicular end open	Less congestive epididymal pain; granuloma linked to better reversal

KEY POINT

Every recommended method rests on **mucosal cautery**; ligatures and clips alone are not core methods. Division is required; segment excision is optional (**1 cm** is adequate).

ALGORITHM

1. Do PVSA at **8–16 weeks**.
2. If azoospermia or rare non-motile sperm (RNMS, ≤ **100,000/mL**), then stop contraception.
3. If motile sperm, then repeat at 6 months; if still motile, then it is a failure, so repeat the vasectomy.

- **Early failure** is technical, such as a missed vas; **late failure** is recanalisation after a successful PVSA.
- **Sperm granuloma** is expected rather than a complication. Post-vasectomy pain affects about **2%**; if a cord block helps, then denervation or reversal follows.

8 · Orchidectomy and prosthesis

- **Bilateral simple orchidectomy** treats hormone-sensitive metastatic prostate cancer. It is cheaper than androgen deprivation therapy but has no proven survival benefit.
- **Unilateral** orchidectomy is for trauma or non-salvageable torsion, and for orchialgia only after all conservative measures. **Epididymis-sparing** orchidectomy gives a better contour.
- **For orchialgia, use the inguinal route**, because it removes more of the proximal ilioinguinal, iliohypogastric and genitofemoral nerves.
- **Prosthesis**: offer it to every man lacking a testis, before cancer orchidectomy. Pierce the fill port ≤ **5 times**, never after insertion. A **high scrotal subdartos** pouch has the least infection and erosion.

9 · Testicular torsion and fixation

- †**Acute torsion**: if suspicion is high, explore without waiting for imaging. Detorse and assess, then fix or remove; **always fix the contralateral testis**.
- Intermittent torsion (pain <2 h) and postpubertal retractile testis need elective orchidopexy.

Fixation	Method	Note
Suture fixation	Albuginea to dartos (3-0 absorbable)	Re-torsion still possible; transparenchymal sutures damage spermatogenesis
Dartos pouch	Subdartos pouch with a purse-string at the cremaster exit	No transparenchymal suture, so **preferred when fertility matters**

10 · Chronic scrotal pain (>3 months)

ALGORITHM

1. History, examination, duplex ultrasound; if a cause is found, then treat it.
2. If none, then exclude extrascrotal causes (abdomen, pelvis, spine, hip, prostatitis, pelvic floor).
3. If non-specific, then give support, NSAIDs, tricyclics or gabapentin for ≥3 months.
4. If epididymal, then epididymectomy.
5. If testicular or post-vasectomy, then do a cord block: if relief >**50%**, then microsurgical denervation (or reversal); if <50%, then pain clinic, with inguinal orchidectomy last.
6. If post-hernia, then ilioinguinal block; if relief >**90%**, then neurolysis.

- **Microsurgical denervation of the spermatic cord (MDSC)** is done through a subinguinal incision. A micro-Doppler helps **preserve testicular and cremasteric arteries, lymphatics, and the vas with its vessels**; everything else, including the trifecta nerve complex, is divided.

11 · Complications of scrotal surgery

- **Haematoma** is commonest; drain it if large, infected or enlarging. A falling haematocrit after orchidectomy suggests a slipped cord ligature with a **retroperitoneal bleed**, which needs re-ligation through a retroperitoneal extension.

12 · Seminal vesicles

- The seminal vesicles (SV) are Wolffian derivatives that bud at about **12 weeks** and make about **60%** of seminal fluid. The **neurovascular bundles lie lateral to their tips**.
- **MRI is most accurate**. On transrectal ultrasound (TRUS), suspect obstruction if the SV is >**1.5 cm** or the ejaculatory duct >**2.3 mm**; confirm by **aspiration showing sperm**.
- **Zinner syndrome** is an SV cyst with **ipsilateral renal agenesis** and ejaculatory duct obstruction (EDO), often with an ectopic ureter into the cyst.
- Observe or aspirate small cysts. **Transurethral unroofing** is preferred for cysts, EDO and stones; large symptomatic cysts need laparoscopic or robotic excision.

Benign	Malignant
Cystadenoma, papillary adenoma, hydatid cyst, amyloid	Adenocarcinoma, mixed epithelial–stromal, phyllodes, sarcoma, carcinoid, seminoma, yolk sac

- Image tumours with MRI and biopsy by TRUS if inconclusive. Secondary invasion is far commoner. Adenocarcinoma needs radical excision and lymphadenectomy.

13 · Surgical approaches to the seminal vesicles

Approach	Use	Main risk
Transvesical transtrigonal	Large benign cyst; ectopic ureter into cyst	Urinary leak, ureter
Perivesical	Children, large cyst with or without nephroureterectomy	Ureter, vas
Transperitoneal (laparoscopic/robotic)	Bilateral lesions; dominant, because it opens neither organ	Access injuries
Perineal	At perineal prostatectomy	Neurovascular bundle
Transcoccygeal	Hostile abdomen/perineum	Rectum

- A rectal injury needs two-layer repair; a ureteric injury needs reimplantation over a stent.

★ Putting it together

A · Testicular mass: the first week

ALGORITHM

1. Ultrasound.
2. Tumour markers AFP, hCG and LDH **before** orchidectomy; sperm bank.
3. **Radical inguinal orchidectomy** (never trans-scrotal).
4. Repeat markers (AFP half-life 5–7 days, hCG 1–1.5 days); CT chest, abdomen and pelvis.

B · Germ cell tumour by stage

Stage	Seminoma	Non-seminoma (NSGCT)
CS I	Surveillance (occult spread 10–15%); single-dose carboplatin	Low risk: surveillance. High risk (lymphovascular invasion, embryonal-predominant): surveillance, BEP ×1 or nerve-sparing RPLND
II	≤3 cm: radiotherapy, chemo or RPLND; >3 cm: chemo	Normal markers and small nodes: RPLND; otherwise chemo
Advanced (IGCCCG)	Full-dose cisplatin. Good: BEP ×3 or EP ×4. Intermediate/poor: BEP ×4 (VIP ×4 if lung concern).	
Residual mass	>3 cm: FDG-PET (≥6 weeks post-chemotherapy)	Any >1 cm: post-chemotherapy RPLND (teratoma)

C · Penile cancer: primary and groin

ALGORITHM

1. **Primary:** if PeIN, then topical 5-FU/imiquimod, laser or glans resurfacing. If invasive, then organ-sparing surgery, glansectomy, or partial/total penectomy; margins <1 cm are acceptable.
2. **cN0:** if low risk (pTis/pTa/pT1a), then surveillance; if intermediate/high, then DSNB or modified ILND; if positive, then radical.
3. **cN+:** if FNA positive, then bilateral radical ILND; if cN3, then neoadjuvant TIP first. Add pelvic dissection if ≥2 inguinal nodes or extranodal extension.

D · Anterior urethral stricture

Stricture	Treatment
Bulbar <1.5 cm, first time	Dilation or DVIU once; if it recurs, urethroplasty
Bulbar 1–2 cm (3–5 cm proximally)	EPA
Longer bulbar	BMG onlay or augmented anastomosis
Penile	Onlay graft/flap; staged if hostile bed, LS, failed hypospadias
Panurethral, frail	Perineal urethrostomy
Pelvic fracture urethral injury	Suprapubic catheter, then delayed perineal anastomosis

In retention, place a suprapubic catheter; never blind-dilate.